

Beyond the Honeymoon Phase: Ordered Stages and Divergent Trajectories in Human-Agent Relationships

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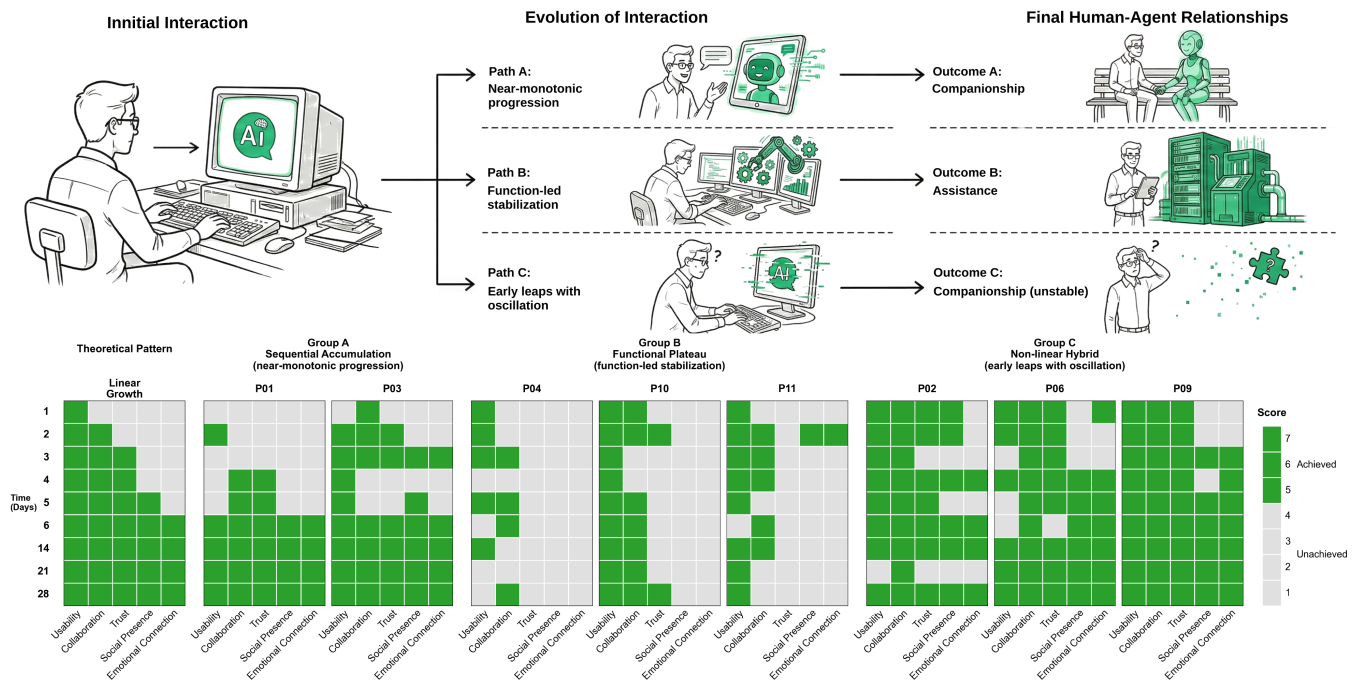


Figure 1: We contrast a hypothesized stage-based development pattern with empirically observed trajectories, showing that participants followed distinct and non-linear progression patterns in a longitudinal case study. Participants are labeled P01–P12; all participants completed Phase 1, and 8 continued to Phase 2.

Abstract

Interpersonal relationships are typically viewed as evolving through identifiable stages that unfold over time. A critical yet unresolved question is whether long-term human-agent relationships parallel the structured developmental trajectories observed in interpersonal

communication. We report a two-phase, four-week longitudinal case study of sustained interaction with ChatGPT, integrating behavioral logs, repeated questionnaires, and in-depth interviews. Guided by interpersonal relationship theory, we operationalized relationship development along five evaluative stages and examined their temporal ordering. Our findings provide converging evidence that while relationship development generally adheres to a hierarchical stage order, the progression is inherently non-monotonic and diverges into three distinct trajectories: sequential accumulation, functional plateau, and non-linear hybrid trajectories. Further, stage transitions hinge on continuity and interaction dynamics rather than interaction frequency alone. These insights advocate

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for "relationship-aware" design paradigms that can dynamically diagnose interaction states and facilitate targeted transitions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Human–AI Relationship, Longitudinal Case Study, Conversational Agents, Relationship Trajectories, Trust in AI, Relationship-Aware Design

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1 Introduction

Human relationships are widely understood as developing through recognizable stages rather than forming instantaneously [4, 16]. Classic interpersonal communication theories, such as Knapp's five-stage relational model, describe how relationships progress from initial contact to deeper emotional bonding through shifts in interaction goals, trust, and mutual understanding [11, 12]. These models emphasize that development depends not only on interaction frequency, but on how people interpret roles, expectations, and emotional boundaries over time.

As conversational agents increasingly enter everyday life, an important question arises: do human–agent relationships follow similar multi-stage patterns, and if so, how do these trajectories unfold in real-world use? Prior work has characterized AI companionship along functional and emotional dimensions [2, 21, 26, 27, 29], yet most empirical studies focus on short-term or task-specific interactions [5, 6, 13, 15, 19]. In practice, emotional engagement is often lightweight and situational rather than sustained, raising questions about whether long-term interaction leads to social reframing, functional stabilization, or distinct relationship trajectories. Understanding these dynamics is critical not only for theory, but also for the design of relationship-aware agent systems. Such systems must adapt beyond immediate tasks to users' evolving relationships, adjusting social roles based on interaction history and changing relational expectations. However, in the absence of longitudinal evidence, current design approaches risk extrapolating human–human relationship models to human–agent interaction without adequate empirical validation.

1.1 Social Role Interpretation in Human–Agent Interaction

Beyond functional performance, users often interpret conversational agents through socially meaningful roles such as assistant, advisor, or emotional supporter [1, 8, 9, 24]. Prior work shows that such role framings shape how users evaluate and rely on agents over time [18, 20]. Liao further distinguishes between sociotechnical relationships, where agents are treated as tools evaluated by

efficiency, and therapeutic relationships, where agents are experienced as emotionally supportive partners characterized by trust and perceived understanding [17]. This distinction suggests that relationship orientation reflects not only system capabilities or tasks, but also users' interpretive and motivational processes, motivating longitudinal investigation of how such framings emerge and stabilize in everyday use.

1.2 Human–Agent Relationships as Dynamic and Multi-Stage Processes

Drawing on Knapp's five-stage relational model as a theoretical foundation and adapting it to human–agent interaction, we operationalize relationship development as five evaluative stages, each associated with a dominant construct (see Figure S1): (1) Instrumental tool-use, assessed primarily by usability; (2) Reliable assistance, characterized by growing sense of collaboration; (3) Collaborative partnership, where trust in the agent becomes central; (4) Social companionship, marked by increased social presence; and (5) Intimate companionship, defined by stronger emotional connection.

1.3 Research Gap and Our Contribution

Staged human-agent relationship development has largely remained a theoretical proposition rather than being systematically tested in naturalistic longitudinal settings. To address this gap, we conduct a four-week longitudinal case study of sustained interaction with a general-purpose conversational agent, integrating behavioral logs, repeated questionnaires, and in-depth interviews to examine how relationship evaluations and usage practices evolve over time.

Contributions. (1) *Theoretical*: We adapt interpersonal stage theory to model evaluative development in human–agent relationships. (2) *Empirical*: We identify three relationship trajectories—sequential accumulation, functional plateau, and non-linear hybrid—based on long-term deployment data. (3) *Design*: We derive diagnostic cues for identifying relationship states and supporting targeted stage transitions in relationship-aware agent systems.

2 Case Study Description

This study investigates how users' perceived relationships with an AI agent evolve over time, transitioning from instrumental task support to socially and emotionally grounded engagement. We adopted a two-phase, four-week longitudinal study design, with structured task-driven interaction in Phase 1 and unconstrained everyday use in Phase 2, enabling comparison between early framing and later relationship evolution.

2.1 Participants

A total of 12 participants were recruited (9 females, 3 males), with a mean age of 25.58 years ($SD = 3.99$; range: 21–35). The sample included an equal number of enterprise employees and university students (6 each). Most participants had only basic prior experience with ChatGPT and no exposure to advanced features (see details in Table S1).

2.2 Study Design and Procedure

2.2.1 Phase 1: Structured Daily Task-Based Case Study (1 Week).

Participants were instructed to use ChatGPT as their primary AI agent and to complete assigned daily tasks following specific interaction rules. Each day introduced new categories of allowable interaction types, while previously unlocked types remained available, forming a cumulative interaction space. Phase 1 was structured with task assistance-oriented tasks on Days 1–4, followed by emotional support-oriented interactions on Days 5–6 (see details in Table S5). Two rounds of semi-structured interviews were conducted on Day 5 and Day 7, marking key transition points in the study.

2.2.2 Phase 2: Longitudinal Case-Based Field Deployment (3 Weeks).

Participants were encouraged to use ChatGPT freely in their daily lives without assigned tasks or interaction constraints. Each week, we conducted interviews to capture changes in usage patterns, perceived agency, trust, emotional attachment, and reflections on newly introduced features (e.g., holiday-released personalized video messages and annual summary). Short weekly questionnaires complemented the interviews by capturing longitudinal shifts in key evaluative constructs (e.g., usability), along with participants' recent mood and personal goals.

2.3 Data Collection and Analysis

We collected three complementary data sources. All chat interactions were logged with timestamps and message content to analyze interaction frequency, temporal distribution, and prompt types (task assistance vs. emotional support). Questionnaires assessed key evaluative constructs (e.g., usability), recent mood, and personal goals, along with standardized individual difference measures (e.g., Big Five Inventory) to support cross-trajectory comparisons (see Supplementary Tables S1–S4) [3, 7, 10, 14, 22, 23, 25, 28]. In addition, interviews were transcribed and analyzed using both thematic and semantic approaches; behavioral, quantitative, and qualitative data were then integrated for temporal and cross-case analyses of relationship development.

Use of AI Systems. We used a generative image model to assist in creating illustrative figures for this paper. These tools were used solely for visual generation and layout exploration and did not contribute to data analysis, interpretation, or writing of the manuscript.

3 Findings

3.1 Do Human-Agent Relationships Follow a Five-Stage Development Process?

We examined whether participants' relationships with the agent followed a hypothesized five-stage sequence adapted from human-human relationship theories: using stage-attainment heatmaps (stage attainment defined as a mean construct rating ≥ 5 on a 7-point Likert scale, reflecting a substantively positive endorsement above the neutral midpoint of 4; exploratory checks using adjacent thresholds yielded directionally consistent patterns), we first tested an in-order (prefix) constraint: within any single measurement, attainment of a later stage should imply attainment of all

earlier stages. Across all measurements, this constraint was satisfied in 78.1% of cases (75/96), indicating that later-stage experiences usually co-occurred with earlier-stage foundations. However, the remaining 21.9% (21/96) violations were not random: 57.1% (12/21) clustered around two systematic breakpoints—*collaboration without usability* (6/21) and *emotional connection without social presence* (6/21). Moreover, most violations occurred early in deployment Day 1–6 (19/21), while later measurements showed markedly fewer order breaks, suggesting increasing order-consistency over time despite non-monotonic fluctuations.

Comparing observed attainment with the Phase 1 schedule, only 6/48 participant×day cells were strictly attained, while most showed early (16/48) or failed (26/48) attainment. These results indicate that the five stages function more as a dependency structure than a monotonic, day-by-day progression. Interview data helps account for these deviations: increased emotional prompting on Days 5–6 did not consistently yield more positive or bonding-like interaction quality. Sentiment analysis showed that emotional-support exchanges were only mildly positive overall (52.2% positive, 40.0% negative). For example, Participant 01 described Day 5 interactions as “*like customer service... polite but distant... lacking sustained emotional exchange*,” making companionship difficult to endorse. Conversely, early attainment was often driven by strong capability cues or partner framing. P10 described Day 1 interactions as working with “*a strong work partner who can quickly summarize and structure things*,” supporting early collaboration and trust, while P02 reported that the agent's replies were “*in-depth, logical, and reliable*,” leading to early trust ratings.

Taken together, the five stages provide a meaningful map of how users evaluate and narrate human-agent relationships, and stage order is usually respected. Yet real trajectories are non-monotonic: higher-stage experiences may appear early or intermittently, and sustained progression toward social or intimate companionship depends on interaction quality and continuity rather than time alone.

3.2 What Relationship Types Emerged Across Participants?

Based primarily on stage-attainment trajectories and triangulated with interviews and log statistics, we grouped participants into **three relationship types**, each showing distinct progression tempo and stability. These groups show that development is not only staged but also typed, differing in progression depth and stability.

Group A — Sequential Accumulation (near-monotonic progression).

Participants in this group (P01, P03) largely followed the hypothesized direction: later-stage attainment typically appeared after earlier stages and remained relatively stable once reached. They also showed sustained engagement in Phase 2 (e.g., total interactions: P01 = 80, P03 = 212). Qualitatively, P03 characterized the relationship as becoming “*more intimate... between assistant and partner, but still task-oriented*,” indicating gradual boundary negotiation rather than abrupt role shifts. This accumulation pattern is consistent with later-phase consolidation in which all emotional-support exchanges in Phase 2 Week 3 were positive.

Group B — Functional Plateau (function-led stabilization). This group (P04, P10, P11) consolidated early-stage constructs (usability and collaboration) but showed limited or delayed expansion into social or emotional stages. Stage attainment tended to stabilize rather than climb. Behavioral logs showed high overall usage but persistently low emotional ratios (e.g., Phase 2 emotional ratio: P04 = 1.2%, P10 = 0, P11 = 11.5%). Participants consistently framed the agent as efficient and helpful but not emotionally engaging. P04 summarized the relationship as “*efficient, agile, pleasant*,” and described it as somewhere “*between assistant and partner*,” but not as companionship. This plateau aligns with the sentiment pattern of tool-dominant segments: tool-use content made up most labeled segments (Tool Usage = 61%).

Group C — Non-linear Hybrid (early leaps with oscillation). Participants in this group (P02, P06, P09) often showed early leaps into trust or emotional connection, followed by plateaus or regressions. For example, P02 reported that social presence in Phase 2 Week 1 felt lower than on Phase 1 Day 6, despite continued trust in task contexts. This reflects differentiation between instrumental trust and social bonding: trust could remain high while perceived companionship declined when expectations rose or interaction quality fluctuated.

3.3 What Factors Shaped Stage Progression and Type Differences?

Across quantitative correlations and interview data, three recurring patterns were observed in relation to relationship trajectories, and their influence varied across relationship types.

Emotional stability is associated with accumulation of social and emotional stages. Quantitatively, emotional stability showed consistently positive associations with both Social Presence and Emotional Connection across repeated measurements ($r = 0.70 \sim 0.87$), indicating that participants with higher emotional stability tended to report stronger and more sustained later-stage evaluations. Participants who sustained higher social presence and emotional connection tended to regulate frustration and tolerate imperfections. In Group A, P01 described selectively engaging with tasks where the agent performed well and lowering expectations elsewhere, which “*reduced frustration and made the interaction feel more companion-like*.”

Failure sensitivity and self-efficacy orientation are associated with functional framing. Quantitative analysis further indicated negative associations between Social Presence and Self-Efficacy ($r = -0.76 \sim -0.81$), suggesting that participants with stronger self-efficacy tended to maintain a functional, tool-like framing of the agent. A similar negative association emerged between Social Presence and failure sensitivity ($r = -0.72 \sim -0.81$), consistent with a predominantly instrumental framing. In Group B, participants emphasized productivity and efficiency, and rarely engaged in emotional dialogue, consistent with low emotional ratios in logs.

Cognitive reappraisal is associated with early collaborative or trust gains. Cognitive Reappraisal was positively associated with both Usability and Collaboration across measurements ($r = 0.74 \sim 0.91$), supporting the interpretation that participants high in reappraisal

tended to reinterpret system limitations as opportunities for joint problem solving. Some participants reframed system limitations and focused on strengths, allowing early gains in collaboration and trust even under suboptimal usability. P02 explicitly described judging the agent by reasoning quality rather than latency or interface friction, enabling early-stage “leaps” but also making later disappointment more salient when social expectations increased. This mechanism was most evident in Group C’s oscillating trajectories.

Across groups, **memory and continuity emerged as prerequisites for sustained progression.** P02 noted that when the agent recalled prior context and asked follow-up questions, it felt more like a “*psychological counselor*,” supporting both trust and emotional connection. In contrast, lack of continuity often collapsed perceived social presence even when task performance remained strong. Overall, personality-linked regulation and expectation management shaped whether relationships accumulated, plateaued, or oscillated, producing distinct long-term types rather than a single universal path.

4 Discussion

The Dynamics of Human-Agent Bonding: Beyond Linear Progression. Our longitudinal analysis challenges the prevailing assumption that human-agent relationship development mirrors a strictly linear escalation. Instead, we propose a type-divergent developmental framework where shared evaluative stages function as a dependency structure rather than a chronological mandate. While the high adherence to stage order (78.1%) confirms a foundational logic similar to Knapp’s model, the observed “order breaks” point to cognitive reappraisal allowing system intelligence to compensate for interface friction and enable early collaborative framing.

From Measurement to Diagnosis: Knowing Where a Product Stands. From a product perspective, relationship stage should not be inferred from time or usage frequency alone, but from patterns of evaluative and behavioral signals (e.g., task success, reliance, reciprocity, and affective exchange). Designers should assess not only which stages are reached, but also their stability and coupling: for instance, trust without social presence may indicate instrumental partnership rather than companionship. Such patterns help determine whether a system supports sequential accumulation, functional stabilization, or oscillating hybrid use.

Designing for Stage Transitions: What Enables the Next Step. To support progression toward more collaborative or relational stages, design should target the mechanisms constraining transition. Moving from usability to collaboration requires shared problem framing (e.g., goal memory and task decomposition), while transitions to trust depend on consistency and transparency in reasoning. Later social and emotional stages rely more on continuity and emotional responsiveness than task performance; improving emotional dialogue quality can lower barriers to re-engagement from instrumental use (see Supplementary Table S7).

Limitations and Future Directions. This study is based on a relatively homogeneous sample ($N = 12$) interacting with a single general-purpose conversational agent, which limits statistical power and generalizability. The reported correlations should therefore be

interpreted as exploratory pattern-level signals rather than confirmatory evidence, particularly given repeated measurements across nine time points, for which no formal correction for multiple comparisons was applied due to the exploratory, pattern-oriented nature of the analysis. The three relationship types should likewise be understood as interpretive trajectory patterns derived from longitudinal cross-case comparison, rather than statistically validated clusters, especially given the small number of participants per group. We emphasize triangulated convergence across behavioral logs, interviews, and temporal trajectories rather than relying on isolated coefficients. Future research with larger and more diverse populations, as well as comparative multi-agent designs across varying personas and system types, is needed to assess the robustness and generalizability of the observed stage dependencies and trajectory patterns.

5 Conclusion

Through a longitudinal case study grounded in interpersonal relationship theory, we show that while users evaluate agents along shared stage dimensions with ordered dependencies, development does not follow a single linear path. Instead, participants diverged into stable relationship types with distinct patterns of accumulation, stabilization, and oscillation. Thus, stages capture what is evaluated, whereas trajectories reflect how evaluations evolve over time, underscoring the need for relationship-aware agent systems that diagnose current states and support targeted transitions toward collaboration or companionship.

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